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4. SPECIAL ENVIRONMENTAL CONSIDERATIONS

4.1 SEWAGE DISPOSAL

The tablelands climate and soil types of Orange influence the effectiveness of on-site sewage management facilities in the City. Sewerage facilities for the Orange urban area were introduced in 1914. Sewerage facilities were also been introduced in the Villages of Lucknow and Spring Hill in the 1980s. The Orange Sewage Treatment Plant has incorporated nutrient-reduction processes to improve the environmental quality of effluent.

Urban development must be connected to a sewerage system approved by Council prior to occupation. For the purposes of this clause, urban development includes all development within residential, business and industrial zones (and associated open space and special use zones) - excluding Clergate Road Industrial Area (CRIA), north of Farrell Road. Industrial development of land in CRIA for dry industries may proceed without sewer provided that the area of the land is sufficient to meet the wastewater demand of industrial development. Sewer has recently been extended to this area. This requirement to connect to sewer also applies to land within urban-transition zones where lots are to be created with an area of less than 2 hectares.

In order to ensure that sewerage facilities continue to have the capacity to service planned urban development areas over the long term, sewerage services are not programmed to rural residential or rural-zoned land unless specific provision has been made in the planned capacity of sewer facilities to service the land. On-site sewage management systems are required for development in rural areas (including rural residential development).

On-site management of wastewater requires that effluent be managed entirely within the subject land.

PO 4.1-1 PLANNING OUTCOMES - SEWAGE DISPOSAL

1 Development within the urban area of Orange as defined above is connected to sewerage facilities or arrangements to the satisfaction of Council have been made for the provision of sewerage services prior to occupation.

2 Where sewerage services are not provided, on-site disposal of effluent is designed and implemented in accordance with the relevant guidelines for on-site sewage management systems.

PO 4.1-1 PLANNING OUTCOMES - SEWAGE DISPOSAL

3 Suitable areas for on-site disposal of effluent are defined prior to:
- issue of a construction certificate for a building; or
- issue of a subdivision certificate for new lots.

4 Lots to be created by subdivision for residential purposes indicate an envelope defining an area suitable for on-site disposal of effluent that has been subject to geotechnical assessment.

5 Rural or rural residential-zoned land is served by on-site sewage management systems unless the land is traversed by sewer mains with adequate planned capacity.

RELATED INFORMATION – SEWAGE DISPOSAL

1 Department of Local Government 1998 *Environment & Health Protection Guidelines: On-site Sewage Management for Single Households.*

2 DLG *Septic safe program:* Septic safe (If it fails to load click on "Septic Safe on the DLG home page)

4.2 DEVELOPMENT NEAR MAJOR INDUSTRY AND UTILITIES

The continued operations of major industry and utilities are important to the growth and future of the City. Some of these activities, however, have impacts that may extend beyond the site boundaries despite all reasonable mitigation measures.

A planning objective for Orange is to protect the continued operation of major development from encroachment of incompatible development. Some of the major developments have been approved subject to acquisition provisions in the event that environmental standards are not met. LEP 2000 identifies areas around major industries and utilities in the City where special development consideration is required for new development. Such major developments are listed below:

Development	**Major Issues**
Sewage Treatment Plants	Odour, noise
"Honeymans Hill" Quarry	Dust, noise, vibration and visual impact
Shadforth Quarry	Dust, noise, vibration and visual impact
Orange Abattoir	Dust, industrial/transport noise and odour
Former Canobolas Wool Topmaking (including wastewater treatment and spray irrigation)	Odour, industrial noise
Orange Airport	Aircraft noise and height limits near flight paths

Development applications in areas affected by clause 14 of LEP 2000 need to include information on measures that are proposed to be taken to ensure that the development is not adversely affected by the continued operations of the major industry or utility installation in the vicinity.

Measures such as perimeter landscaping for screening, buffers or filters, siting and orientation of buildings and sound attenuation in buildings may also be necessary.

For development in the vicinity of Orange Airport, buildings are to be designed and constructed according to ***Australian Standard AS2021-1994 (Acoustics-Aircraft Noise intrusion- Building Siting and Construction*)**. For determining compliance with the above standard, the hatched area around the airport is deemed to comprise the **ANEF25** contour.

PO 4.2-1 PLANNING OUTCOMES - IN THE VICINITY OF MAJOR INDUSTRY OR UTILITY INSTALLATIONS

1 Development comprising new buildings on land in the hatched area around Orange Airport include measures in accordance with ***Australian Standard AS2021-1994 (Acoustics-Aircraft Noise intrusion- Building Siting and Construction*** . Consideration should also be given to this Standard for uses in the vicinity.

PO 4.2-1 PLANNING OUTCOMES - IN THE VICINITY OF MAJOR INDUSTRY OR UTILITY INSTALLATIONS

2 Development applications indicate measures proposed with development on land within the hatched area to minimise associated impacts.

3 Development on land affected by the hatched area around major industries and utilities incorporates effective measures to minimise potential impacts from those industries or utilities.

RELATED INFORMATION – DEVELOPMENT NEAR MAJOR INDUSTRY AND UTILITIES

1 ***Australian Standard AS2021-1994 (Acoustics-Aircraft Noise intrusion- Building Siting and Construction***

2 Department of Planning ***Circular No E3; Guidelines for buffer areas around sewage treatment (water pollution control) plants***

4.3 LAND SHAPING

Substantial land shaping, including filling or excavating land, has the potential to generate significant environmental impacts. Land shaping should not adversely affect land in the vicinity as a result of drainage, erosion or sedimentation or as a result of leachate entering surface or groundwater.

Development applications are required for land shaping unless it forms part of a development that is carried out in accordance with a development consent.

Development consent is not required for land shaping which, in the opinion of Council, is minor.

Minor land shaping are deemed to include work that involves changes in elevation of less than 600mm on land that is located outside a water-quality-protection area or at least 40 metres from any defined waterway.

Minor land shaping does not comprise any work undertaken on land previously disturbed by mining activities in and around the Village of Lucknow. Land shaping within areas disturbed by mining activity in and around the Village of Lucknow requires development consent.

Land shaping within a water-quality-protection area or within 40 metres of a defined waterway (creek or water storage) requires development consent.

Land that is likely to be contaminated requires development consent for land shaping (refer to Section 4.4 below for contaminated-land issues).

All land shaping, whether or not consent is also required, are to incorporate soil-erosion control measures. Works are to be carried out in a way that does not prevent effective drainage on or from the land and does not concentrate flow onto adjoining land without incorporating proper drainage measures (including the creation of easements as required).

Where development consent is required, a soil-erosion management plan should be submitted with the development application.

Land shaping comprising soil-conservation works does not require development consent but may involve independent environmental assessment by the Department of Infrastructure, Planning and Natural Resources (formerly DLWC) under Part 5 of the Act.

PO 4.3-1 PLANNING OUTCOMES - LAND SHAPING

1 Applications include details on the existing and proposed landform, watercourses and vegetation.

PO 4.3-1 PLANNING OUTCOMES - LAND SHAPING

2 Applications are accompanied by a soil-erosion control plan for implementation prior to and upon commencement of the work.

3 Landfilling comprises inert material only and does not include putrescible waste, vegetation, or other material that may decompose.

4 Landfill is compacted to the required standard and evidence of compaction rates are provided upon completion of the work or otherwise as directed by Council.

RELATED INFORMATION – LAND SHAPING

DIPNR: Soil Conservation

4.4 CONTAMINATED LAND

Contaminated land can pose a risk to human health and the environment. Contaminants can be released into waterways and humans exposed when contaminated land is disturbed as a consequence of development.

Determining how contaminated land can be used for different purposes involves a three-stage response: identification; evaluation; and clean-up.

- Identification involves determining if a site is potentially contaminated.
- Evaluation defines the nature and extent of contamination by sampling and analysis.
- Clean-up involves selecting a remediation method, implementing and validating remediation and issuing post-remediation clearances.

The Table on the following page indicates how different land use in Orange may typically be potentially contaminated. The Table provides a guide only to possible contaminants. Others may also be present.

PO 4.4-1 PLANNING OUTCOMES FOR CONTAMINATED LAND

1 Land subject to development is clear from contamination.

2 Development complies with the Contaminated Land Management Act 1997.

3 Applications for development consent on land used or likely to have been previously used for uses in the table below include contamination assessment and where necessary a proposed remediation strategy to make the site suitable for the proposed use.

4 An independent site audit at the applicant's cost is carried out to assess the information provided with an application where Council considers that:
- information may be incorrect or incomplete;
- it needs to verify that the information adheres to appropriate standards, procedures or guidelines; or
- the type or level of contamination requires an independent technical review.

RELATED INFORMATION – CONTAMINATED LAND

1 Environment Protection Authority, 1995. ***Sampling Design Guidelines***

2 Department of Urban Affairs and Planning & EPA, 1995***. Contaminated Land: Planning Guidelines for Contaminated Land*** - superseded by the 1998 draft guidelines.

3 Department of Urban Affairs and Planning & EPA, 1998 ***Managing Land Contamination: Draft Planning Guidelines***

4 Orange City Council ***State of Environment Reports***

5 EPA overview of contaminated land management: EPA Overview

6 EPA Contaminated Land Management Guidelines

7 State Environmental Planning Policy – ***Remediation of Land***

Land Use	Example	Main Contaminants*
Agriculture	1 Land heavily treated with chemicals including land used for orchards and other horticulture, stock dips, chemical mixing or storage 2 Areas where fuels have been used such as machinery sheds, fuel storage and refuelling areas 3 Areas used for the dumping of waste material	Organochlorins and/or organophosphate chemicals, heavy metals such as arsenic and mercury. Hydrocarbons, oils and heavy metals Hydrocarbons, oils and heavy metals
Airports		Hydrocarbon fuels and solvents, oils and heavy metals
Chemical works including laundries, dry-cleaning establishments, fertiliser-mixing works	Works used in production, recovery and storage of chemicals, including petrochemicals (ie, tar and bitumen), dry-cleaning fluids, fertilisers, pesticides, detergents, inks, paints or dyes, acids and bases.	Organic and inorganic chemicals, petrochemicals, solvents such as chloroform, trichloroethylene, etc
Concrete or brick works	Concrete plants and brick works	Waste used in filling of open cut Hydrocarbons, oils
Gasworks	Gas-generating and storage plants	Hydrocarbons, iron and calcium oxides, asbestos
Heavy-engineering installations	Electrical manufacturing	Heavy metals, oxides, lead, asbestos.
Metal works	Metal recovery, electroplating and metal finishing	Phosphates, nitrates, nitrites, sulphates, heavy metals, aromatic compounds and chlorinated hydrocarbons, PCBs, mineral oils, asbestos
Mines and extractive industries	Sites used for mining and processing of metals and quarries.	Acids, cyanides and heavy metals, TNT, nitrates and nitrates, detonating devices, hydrocarbons, various wastes used to fill in shafts and open cut
Pesticide-storage areas		Insecticides, fungicides, herbicides
Petroleum storage and distribution	• Fuel depots • Service stations • Above-ground and underground storage tanks	Petroleum hydrocarbons, oils and lead
Premises using radioactive materials	Research and medical establishments	Radioactive isotopes, cobalt, strontium, caesium, radium, uranium, thorium, plutonium and tritium
Railway land	Sidings and depots	Oils and hydrocarbons, volatile chlorinated hydrocarbons, heavy metals, arsenic, herbicides, asbestos
Scrap-metal yards		Heavy metals, residues from metal containers
Termite/ant control	Where substantial areas of soil are affected	Organochlorins
Timber-treatment works		Creosote, hydrocarbons, copper, chromium, arsenic, various wood preservatives and organochlorins
Waste disposal. Areas (including transfer stations	• Solid waste landfill, • Spray irrigation of industrial wastewater	Various domestic and industrial wastes Chemicals used or released in processing

4.5 SCENIC AREAS

Buildings in scenic areas should blend into the landscape so as not to be visually intrusive.

Buildings should be constructed using predominantly non-reflective external materials. Muted shades of greens and earth tones should predominate so that buildings blend into the landscape when viewed from roads, public areas and residential areas. Shades of white, primary colours (red, yellow and blue) or other bright colours should be used with restraint to supplement the dominant natural tones.

Particular attention should be given to roofing colours due to the visual dominance of the roof form. As a guide, colours and shades that may be appropriate in rural areas (based on standard "Colorbond" ® colours) are indicated below:

Rivergum (**C**), Caulfield Green (**C**), Slate Grey Mist Green (**C**), Saltbush (**C**).

The use of Zincalume roofs may be appropriate in heritage settings or against skylines, however, the potential generation of glare needs to be considered.



Photograph - Rural dwelling framed by trees and using appropriate colours

PO 4.5-1 PLANNING OUTCOMES FOR SCENIC AREAS

1 Development is carried out in a way that retains, embellishes or replicates a natural, landscaped setting. Buildings are framed by trees or set below the ridgeline. Ridgelines are dominated by trees, preferably native trees, although shelter belts and hedges may be appropriate in some locations.

2 Trees that contribute to the landscape character are retained and embellished in scenic areas.

3 The external materials of buildings reflect the natural colours of the settings – ie, with shades of natural green and browns predominating over supplementary brighter colours.

4 Details of proposed colour schemes, finishes of external materials, existing trees and proposed landscaping are submitted for Council endorsement prior to the issue of a construction certificate.

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